

TECHNICAL & INSTALLATION REFERENCE

Titan Premium Series

Variable Capacity Split-System Heat Pump & Air Handler

Models Covered: TPH-024-V, TPH-036-V, TPH-048-V, TAH-2436-C

Refrigerant Typology: EcoShield R-32 / R-410A High-Pressure Loops

Document ID: TSM-Variable-2026-X4

Release Date: June 2026

1. General Safety Boundaries & Regulations

Installation, commissioning, and maintenance of the Titan Premium Series split-system must only be executed by fully certified, licensed HVAC professionals. Unqualified handling can result in critical component failure, severe property damage, personal injury, or electrocution.

CRITICAL WARNING: HIGH VOLTAGE & PRESSURE

Always verify that all primary power disconnect switches are locked out in the **OFF** position before attempting physical access or wiring modifications. The variable-capacity inverter module stores lethal residual voltage in its capacitor banks for up to 10 minutes after main supply interruption.

Never ground the indoor or outdoor physical chassis to gas supply lines, plastic water pipelines, or telephone signal carriers. A dedicated earth-ground termination path is mandatory to prevent intermediate localized tracking faults and structural fire hazards.

2. Structural Placement & Clearances

Correct structural orientation and localized airflow separation guarantees that the variable capacity condenser can reject heat efficiently without encountering air recalculation short-circuits.

2.1 Outdoor Condenser Unit Positioning

The outdoor condenser must be anchored onto an engineered, rigid level mounting surface or structural pad elevated at least 4 inches above the historical localized snow line. Maintain a minimum of **24 inches** of clear, uninhibited service access on the electrical panel face and at least **48 inches** of completely open clearance directly above the fan discharge stack.

2.2 Indoor Air Handler Alignment

If installing the matching air handler inside a finished ceiling space or directly over an active residential living area, building codes mandate the installation of a field-supplied secondary condensate overflow pan equipped with an isolated flood switch loop.

Acoustic and Inverter Signal Isolation

To completely isolate structural and radio frequency (RF) operational interference, route all indoor control wires at least 3.3 feet (1 meter) away from high-frequency lighting ballasts, smart routers, and household appliance lines.

3. Refrigerant Circuiting & Line Set Logic

The Titan Premium Series operates utilizing modern, high-pressure variable flow cycles. All interconnected copper tube line sets must be strictly constructed from specialized refrigerant-grade clean deoxidized copper tubing (ACR).

3.1 Line Set Sizing Matrix

Refer to the technical chart below to define appropriate line diameters based on total equivalent installation distances:

System Capacity Tonnage	Liquid Line Outer Diam.	Vapor Line Outer Diam.	Max Equivalent Line Length	Max Vertical Lift Allowance
2.0 Tons (TPH-024-V)	3/8 in.	3/4 in.	100 ft. (30m)	50 ft. (15m)
3.0 Tons (TPH-036-V)	3/8 in.	7/8 in.	125 ft. (38m)	60 ft. (18m)
4.0 Tons (TPH-048-V)	3/8 in.	7/8 in.	150 ft. (45m)	80 ft. (24m)

3.2 Line Set Evacuation Protocol

If the internal circuit or matching indoor vapor coil has been opened to atmospheric pressure for longer than 5 continuous minutes, a comprehensive deep vacuum sweep must be initiated.

- Nitrogen Pressure Test:** Pressurize the interconnected lineset loops to 450 PSIG using dry nitrogen gas. Maintain this pressure for 60 consecutive minutes to confirm structural flare and weld integrity.
- Triple Evacuation Sweep:** Connect a dual-stage industrial vacuum pump. Evacuate the sealed circuit down to a level below **500 microns**.
- Vacuum Hold Verification:** Isolate the pump valves completely. The absolute system internal pressure must hold below 500 microns for a minimum of 15 minutes without shifting upward, proving zero residual non-condensable atmospheric moisture remains inside the loop.

4. Electrical Interface & Smart Bus Wiring

Main supply power distribution parameters must conform fully with individual system nameplate values and local structural building codes.

CAUTION: SIGNAL INTERRUPTIONS

Do not pass high-voltage AC mains wiring through the identical conduit sleeve housing low-voltage digital signal carriers. This will induce electromagnetic noise on the communication bus lines, triggering random communication loop failures (Fault E7).

4.1 High Voltage Parameters

Verify that the single-phase or three-phase line potential stays bounded safely within the normal operating matrix (+10% / -10% of standard plate ratings). Use copper wire conductors exclusively.

4.2 Intelligent Communicating Control Bus

The Titan variable intelligence module relies on a proprietary 4-wire shielded communication pathway (designated terminal ports A-B-C-D). Use a minimum of 18 AWG shielded twisted-pair conductors for all inter-unit communication links exceeding 50 linear feet.

5. Commissioning & Subcooling Verification

Do not execute formal system subcooling measurements or adjust factory base charge weights if the outside local ambient testing temperature registers under 55°F (13°C).

5.1 Charging Computations

The physical system charge level is accurately verified via the target liquid line subcooling method. To calculate the current true subcooling value, utilize the physical calculation definition:

$$SC = T_{sat} - T_{liquid}$$

Where T_{sat} represents the exact saturation temperature corresponding directly to the liquid line high-pressure manifold gauge pressure reading, and T_{liquid} is the absolute physical temperature recorded directly on the outside exterior surface of the copper liquid pipe.

5.2 Subcooling Adjustments

Operate the variable system on absolute maximum cooling mode for at least 15 continuous minutes until system pressures completely stabilize. Compare your calculated values with the factory targets:

- If the calculated value matches within $\pm 0.6 \text{ oz/ft}$ of standard layout metrics, the total closed factory balance is verified.
- If subcooling is below target threshold parameters, add pure liquid phase refrigerant slowly into the low pressure suction port loop.

6. System Diagnostics & Fault Codes

The integrated control chassis continuously tracks overall equipment status. Active operational system warnings are instantly broadcast via the primary diagnostic alphanumeric screen or communicating thermostat interface:

Code	Identified System Malfunction	Root Trigger Definition	Recommended Corrective Action Path
E1	High Discharge Pressure Trap	Dirty outdoor coil / Blocked air paths	Clean condenser, inspect external fan motor limits.
E7	Control Bus Communication Fault	Interrupted A-B-C-D signal wiring lines	Check terminal loop continuity, inspect shielding grounds.
F3	Discharge Line Overheating	Severely low system refrigerant fluid charge	Perform isolated electronic leak sweep; recharge to subcooling spec.
P4	Inverter Module Heat Sink Fault	Degraded thermal paste / Fan lockup	Clean heat sink face; reapply matching OEM thermal compound layer.